Tacrolimus: There have been reports that an interaction exists when fluconazole is administered concomitantly with tacrolimus, leading to increased serum levels of tacrolimus. There have been reports of nephrotoxicity in patients to whom fluconazole and tacrolimus were co-administered. Patients receiving tacrolimus and fluconazole concomitantly should be carefully monitored.
Zidovudine: Two kinetic studies resulted in increased levels of zidovudine most likely caused by the decreased conversion of zidovudine to its major metabolite. One study determined zidovudine levels in AIDS or ARC patients before and following fluconazole 200 mg daily for 15 days. There was a significant increase in zidovudine AUC (20%). A 2nd randomized, 2-period, 2-treatment cross-over study examined zidovudine levels in HIV infected patients. On 2 occasions, 21 days apart, patients received zidovudine 200 mg every 8 hrs either with or without fluconazole 400 mg daily for 7 days. The AUC of zidovudine significantly increased (74%) during co-administration with fluconazole. Patients receiving this combination should be monitored for the development of zidovudine-related adverse reactions.
The use of fluconazole in patients concurrently taking cisapride, astemizole, rifabutin, tacrolimus or other drugs metabolized by the cytochrome P-450 system may be associated with elevations in serum levels of these drugs. In the absence of definitive information, caution should be used when co-administering fluconazole. Patients should be carefully monitored. (See Precautions.)
Interaction studies have shown that when oral fluconazole is coadministered with food, cimetidine, antacids or following total body irradiation for bone marrow transplantation, no clinically significant impairment of fluconazole absorption occurs.
Physicians should be aware that drug-drug interaction studies with other medications have not been conducted, but such interactions may occur.

RECOMMENDED DOSE:

The daily dose of fluconazole should be based on the nature and severity of the fungal infection. Most cases of vaginal candidiasis respond to single-dose therapy. Therapy for those types of infections requiring multiple-dose treatment should be continued until clinical parameters or laboratory tests indicate that active fungal infection has subsided. An inadequate period of treatment may lead to recurrence of active infection. Patients with AIDS and cryptococcal meningitis or recurrent oropharyngeal candidiasis usually require maintenance therapy to prevent relapse.
Adults: **Cryptococcal Meningitis and Cryptococcal Infections at Other Sites:** Usual Dose: 400 mg on the 1st day followed by 200-400 mg once daily. Duration of treatment for cryptococcal infections will depend on the clinical and mycological response, but is usually at least 6-8 weeks for cryptococcal meningitis.
Prevention of Relapse of Cryptococcal Meningitis in Patients with AIDS: After the patient receives a full course of primary therapy, fluconaole may be administered indefinitely at a daily dose of at least 200 mg.
Candidaemia, Disseminated Candidiasis and Other Invasive Candidal Infections: Usual Dose: 400 mg on the 1st day followed by 200 mg daily. Depending on the clinical response, the dose may be increased to 400 mg daily. Duration of treatment is based upon the clinical response.
Oropharyngeal Candidiasis: Usual Dose: 50 mg once daily for 7-14 days. If necessary, treatment can be continued for longer periods in patients with severely immune function.
Atrophic Oral Candidiasis Associated with Dentures: Usual Dose: 50 mg once daily for 14 days administered concurrently with local antiseptic measures to the denture.
For other candidal infections of the mucosa, (except vaginal candidiasis, see following text, eg esophagitis, non-invasive bronchopulmonary infections, candiduria, mucocutaneous candidiasis, etc) the usual effective dose is 50-100 mg daily, given for 14-30 days.
For the prevention of relapse of oropharyngeal candidiasis in patients with AIDS, after the patient receives a full course of primary therapy, fluconazole may be administered at a 150 mg once weekly dose.
Treatment of Vaginal Candidiasis: 150 mg should be administered as a single oral dose. To reduce the incidence of recurrent vaginal candidiasis, a 150 mg once monthly dose may be used. The duration of therapy should be individualized, but ranges from 4-12 months. Some patients may require a more frequent dosing.
Candida Balanitis: 150 mg should be administered as a single oral dose.

The recommended fluconazole dosage for the prevention of candidiasis is 50-400 mg once daily, based on the patient's risk for developing fungal infection.
Dermal Infections Including Tinea Pedis, Corporis, Cruris and Candida Infections: Recommended dosage: 150 mg once weekly or 50 mg once daily. Duration of treatment is normally 2-4 weeks but tinea pedis may require treatment for up to 6 weeks.
Tinea Versicolor: Recommended Dose: 300 mg once weekly for 2 weeks; a 3rd weekly dose of 300 mg may be needed in some patients, whereas, in some patients, a single dose of 300-400 mg may be sufficient. An alternate dosing regimen is 50 mg once daily for 2-4 weeks.
For patients at high risk of systemic infection, eg patients who are anticipated to have profound or prolonged neutropenia, the recommended daily dose is 400 mg once daily. Fluconazole administration should start several days before the anticipated onset of neutropenia, and continue for 7 days after the neutrophil count rises above 1000 cells/mm^3.
Dermal Infections Including Tinea Pedis, Corporis, Cruris and Candida Infections: Recommended dosage: 150 mg once weekly or 50 mg once daily. Duration of treatment is normally 2-4 weeks but tinea pedis may require treatment for up to 6 weeks.
Tinea Versicolor: Recommended Dose: 300 mg once weekly for 2 weeks; a 3rd weekly dose of 300 mg may be needed in some patients, whereas, in some patients, a single dose of 300-400 mg may be sufficient. An alternate dosing regimen is 50 mg once daily for 2-4 weeks.
Elderly: Where there is no evidence of renal impairment, normal dosage recommendations should be adopted. For patients with renal impairment (creatinine clearance <50 mL/min) the dosage schedule should be adjusted as described in the following.
Patients with Renal Impairment: Fluconazole is predominantly excreted in the urine as unchanged drug. No adjustments in single-dose therapy are necessary. In patients with impaired renal function who will receive multiple doses of fluconazole, an initial loading dose of 50 mg to 400 mg should be given. After the loading dose, the daily dose (according to indication) should be based on the following table:

Creatinine Clearance (mL/min)	**Percent of Recommended Dose**
>50	100%
11-50	50%
Patients receiving regular dialysis	One dose after every dialysis session

SYMPTOMS AND TREATMENT FOR OVERDOSE:

In the event of overdosage, supportive measures and symptomatic treatment, with gastric treatment with gastric lavage, if necessary, may be adequate. Fluconazole is largely excreted in the urine; forced diuresis would probably increase the elimination rate. A 3-hr haemodialysis session decreases plasma levels by approximately 50 %.

PACKING / PACK SIZES:

150 mg: Blister of 1 capsule
100 mg: Blister of 4 × 7's capsules
50 mg: Blister of 2 × 7's capsules

STORAGE CONDITIONS, USER INSTRUCTIONS AND PHARMACEUTICAL PRECAUTIONS:

STORE BELOW 30°C.

SHELF LIFE: 3 Years

NAME AND ADDRESS OF MANUFACTURER AND DISTRIBUTOR IN MALAYSIA:

SM PHARMACEUTICALS SDN BHD (218620-M)
Lot 88, Sungai Petani Industrial Estate
08000 Sungai Petani
Kedah Darul Aman, Malaysia.

18.01.2018

SM PHARMACEUTICALS SDN BHD

ZOCOL CAPSULES
(Fluconazole 150mg/100mg/50mg)

DESCRIPTION:

150 mg: Opaque white / opaque white colored, size 1 hard gelatin capsules
100 mg: Opaque bluish green / opaque white colored, size 2 hardgelatin capsules.
50 mg: Opaque light blue / opaque white colored, size 4 hard gelatin capsules

COMPOSITION:

150 mg: Each capsule contains: Fluconazole 150 mg
100 mg: Each capsule contains: Fluconazole 100 mg
50 mg: Each capsule contains: Fluconazole 50 mg

PHARMACOLOGY:

Fluconazole, a member of a new class of triazole antifungal agents, is a potent and specific inhibitor of fungal sterol synthesis. Fluconazole is highly specific for fungal cytochrome P-450 dependent enzymes. Fluconazole 50 mg daily given up to 28 days has been shown to affects testosterone plasma concentrations in males or steroid concentrations in females of childbearing age. Fluconazole 200-400 mg daily has no clinically significant effect on endogenous steroid levels or on ACTH-stimulated response in healthy male volunteers. Interaction studies with antipyrine indicate that single or multiple doses of fluconazole 50 mg do not affects its metabolism.
Microbiology: Fluconazole was active in a variety of animal fungal infection models. Activity has been demonstrated against opportunistic mycoses, eg infections with Candida sp, including systemic candidiasis in immunocompromised animals: with Crytococcus neoformans, including intracranial infections; with Microsporum sp; and with Trichophyton sp. Fluconazole has also been shown to be active in animal models of endemic mycoses, including infections with Blastomyces dermatitides; with Coccidioides immitis, including intracranial infection; and with Histoplasma capsulatum in normal and immunosuppressed animals. Fluconazole is highly specific for fungal cytochrome P-450 dependent enzymes. Fluconazole 50 mg daily given up to 28 days has been shown not to affect testerone plasma concentrations in males or steroid concentrations in females of childbearing age. Fluconazole 200-400 my daily has no clinically significant effect on endogenous steroid levels or on ACTH stimulated response in healthy male volunteers. Interaction studies with antipyrine indicate that single or multiple doses of fluconazole 50 mg do not affect its metabolism. After oral administration fluconazole is well absorbed, and plasma levels levels (and systemic bioavailability) are >90% of the levels achieved after IV administration. Oral absorption is not affected by concomitant food intake. Peak plasma concentrations in the fasting state occur between 0.5 and 1.5 hrs post-dose with a plasma elimination half-life of approximately 30 hrs. Plasma concentrations are proportional to dose. Ninety percent steady state levels are reached by day 4-5 with multiple once daily dosing. Administration of loading dose (on day 1) of twice the usual daily dose enables plasma levels to approximate to 90% steady-state levels by day 2. The apparent volume of distribution approximates to total body water. Plasma protein binding is low (11-12%). Fluconazole achieves good penetration in all body fluids studied. The levels of fluconazole in saliva and sputum are similar to plasma levels. In patients with fungal meningitis, fluconazole levels in the CSF are approximately 8-% the corresponding plasma levels. High skin concentrations of fluconazole, above serum concentrations, are achieved in the stratum corneum, epidermis-dermis and eccrine sweat. Fluconazole accumulates in the stratum corneum. At a dose of 50 mg once daily, the concentration of fluconazole after 12 days was 73 mcg/g and 7 days after cessation of treatment the concentration was still 5.8mcg/g. at the 150 mg once-a-week dose, the concentration of fluconazole in stratum corneum on day 7 was 23.4 mcg/g and 7 days after the second dose was still 7.1 mcg/g. Concentration of fluconazole in nails after 4 months of the 150 mg once-a-week dosing was 4.05 mcg/g in healthy and 1.8 mcg/g in diseased nails; and, fluconazole was still measurable in nail samples of 6 months after the end of therapy. The major route of excretion is renal, with approximately 80% of the administered dose appearing in the urine as unchanged drug. Fluconazole clearance is proportional to creatinine clearance.

There is no evidence of circulating metabolites. The long plasma elimination half-life provides the basis for single dose therapy for vaginal candidiasis, once daily and once weekly dosing for other indications. A study compared the saliva and plasma concentrations of a single fluconazole 100 mg dose administered in a capsule or in an oral suspension by rinsing and retaining in mouth for 2 min and swallowing. The maximum concentration of fluconazole in saliva after the suspension was observed 5 min after ingestion, and was 182 times higher than the maximum saliva concentration after the capsule, which occurred 4 hrs after ingestion. After about 4 hrs, the saliva concentrations of fluconazole were similar. The mean AUC_{0-96} in saliva was significantly greater after the suspension compared to the capsule. There was no significant difference in the elimination rate from saliva or the plasma pharmacokinetic parameters for the two formulations. The major route of excretion is renal, with approximately 80% of the administered dose appearing in the urine as unchanged drug. Fluconazole clearance is proportional to creatinine clearance. There is no evidence of circulating metabolites. The long plasma elimination half-life provides the basis for single-dose therapy for vaginal candidiasis and once-daily dosing for all other indications.

INDICATIONS:
Therapy may be instituted before the results of the cultures and other laboratory studies are known; however, once these results become available, anti-infective therapy should be adjusted accordingly.
Cryptococcosis, including cryptococcal meningitis and infections of other sites (e.g., pulmonary, cutaneous). Normal hosts and patients with AIDS, organ transplants or other causes of immunosuppression may be treated. Fluconazole can be used as maintenance therapy to prevent relapse of cryptococcal disease in patients with AIDS.
Systemic candidiasis including candidemia, disseminated candidiasis and other forms of invasive candidal infection. These include infections of the peritoneum, endocardium and pulmonary and urinary tracts. Patients with malignancy, in intensive care units, receiving cytotoxic or immunosuppressive therapy, or with other factors predisposing to candidal infection may be treated. Mucosal candidiasis including oropharyngeal, esophageal, non-invasive bronchopulmonary infections, candiduria, mucocutaneous and chronic oral atrophic candidiasis (denture sore mouth). Normal hosts and patients with compromised immune function may be treated.
Prevention of relapse of oropharyngeal candidiasis in patients with AIDS. Genital candidiasis, vaginal candidiasis, acute or recurrent; and prophylaxis to reduce the incidence of recurrent vaginal candidiasis (=3 episodes a year). Candidal balanitis.
Prevention of fungal infections in patients with malignancy who are predisposed to such infections as a result of cytotoxic chemotherapy or radiotherapy. Dermatomycosis including tinea pedis, tinea corporis, tinea cruris, pityriasis versicolor and candida infections.

SIDE EFFECTS:
Fluconazole is generally well tolerated. The commonest side effects associated with fluconazole are symptoms associated with the gastrointestinal tract. These include nausea, abdominal discomfort, diarrhoea and flatulence. After gastrointestinal symptoms, the second most commonly observed side effect was rash. Headache has been associated with fluconazole. In some patients, particularly those with serious underlying diseases, e.g. AIDS and cancer, changes in renal and hematological function test results and hepatic abnormalities have been observed during treatment with fluconazole and comparative agents, but the clinical significance and relationship to treatment is uncertain. Exfoliative skin disorders, seizures, leukopenia including neutropenia and agranulocytosis, thrombocytopenia, and alopecia have occurred under conditions where a causal association is uncertain. If a rash, which is considered attributable to fluconazole develops in a patient treated for a superficial fungal infection, further therapy with this agent should be discontinued. If patients with invasive/ systemic fungal infections develop rashes, they should be monitored closely and fluconazole discontinued if bullous lesions or erythema multiforme develop.
In rare cases, as with other azoles, anaphylaxis has been reported.

PRECAUTIONS / WARNINGS:
Fluconazole has been associated with rare cases of serious hepatic toxicity including fatalities, primarily in patients with serious underlying medical conditions.

In cases of fluconazole-associated hepatotoxicity, no obvious relationship to total daily dose, duration of therapy, sex or age of patient has been observed. Fluconazole hepatotoxicity has usually been reversible on discontinuation of therapy. Patients who develop abnormal liver function tests during fluconazole therapy should be monitored for the development of more serious hepatic injury. Fluconazole should be discontinued if clinical signs or symptoms consistent with liver disease develop that may be attributable to fluconazole. Patients have rarely developed exfoliative cutaneous reactions, e.g. Stevens-Johnson syndrome and toxic epidermal necrolysis, during treatment with fluconazole. AIDS patients are more prone to the development of severe cutaneous reactions to many drugs. If a rash, which is considered attributable to fluconazole, develops in a patient treated for a superficial fungal infection, further therapy with this agent should be discontinued. If patients with invasive/ systemic fungal infections develop rashes, they should be monitored closely and fluconazole discontinued if bullous lesions or erythema multiforme develop.

The co-administration of fluconazole at doses <400 mg/ day with terfenadine should be carefully monitored. There have been reports of cardiac events including torsade de pointes in patients receiving concomitant administration of fluconazole with cisapride. Patients should be carefully monitored if fluconazole is to be co-administered with cisapride. In rare cases, as with other azoles, anaphylaxis has been reported.

Effects on the Ability to Drive or Operate Machinery: Experience with fluconazole indicates that therapy is unlikely to impair a patient's ability to drive or use machinery.
Carcinogenicity: Fluconazole showed no evidence of carcinogenic potential in mice and rats treated orally for 24 months at doses of 2.5, 5 or 10 mg/kg/day (approximately 2-7 times the recommended human dose). Male rats treated with 5 and 10 mg/ kg/ day had an increased incidence of hepatocellular adenomas.
Mutagenicity: Fluconazole, with or without metabolic activation, was negative in tests for mutagenicity in 4 strains of *S. typhimurium*, and in the mouse lymphoma L5178Y system. Cytogenetic studies *in vivo* (murine bone marrow cells, following oral administration of fluconazole) and *in vitro* (human lymphocytes exposed to fluconazole at 1000 mcg/mL) showed no evidence of chromosomal mutations.
Impairment of Fertility: Fluconazole did not affect the fertility of male or female rats treated orally with daily doses of 5, 10 or 20 mg/kg or with parenteral doses of 5, 25 or 75 mg/kg, although the onset of parturition was slightly delayed at 20 mg/ kg orally. In an intravenous perinatal study in rats at 5, 20 and 40 mg/kg, dystocia and prolongation of parturition were observed in a few dams at 20 mg/kg (approximately 5-15 times the recommended human dose) and 40 mg/ kg, but not at 5 mg/ kg. The disturbances in parturition were reflected by a slight increase in the number of still born pups and decrease of neonatal survival at these dose levels. The effects on parturition in rats are consistent with the species specific estrogen-lowering property produced by high doses of fluconazole. Such a hormone change has not been observed in women treated with fluconazole.

PREGNANCY AND LACTATION:
Use in pregnancy: There are no adequate and well-controlled studies in pregnant women. Adverse fetal effects have been seen in animals only at high dose levels associated with maternal toxicity. These findings are not considered relevant to fluconazole used at therapeutic doses. Use in pregnancy should be avoided except in patients with severe or potentially life-threatening fungal infections in whom fluconazole may be used if the anticipated benefit outweighs the possible risk to the fetus.

There have been reports of spontaneous abortion and congenital abnormalities in infants whose mothers were treated with 150mg of flucanozole as a single or repeated dose in the first trimester.
Use in pregnancy should be avoided except in patients with severe or potentially life-threatening fungal infection in whom Zocol may be used if the anticipated benefit outweighs the possible risk to the fetus. If this drug is used during pregnancy, or if the patient becomes pregnant while taking the drug, the patient should be informed of the potential hazard to the fetus.
Effective contraceptive measures should be considered in women of child-bearing potential and should continue thought the treatment period and for approximately 1 week (5 to 6 half-lives) after the final dose.
There have been reports of multiple congenital abnormalities in infants whose mothers were treated with high-dose (400mg/ day to 800mg/ day) flucanozole therapy for coccidioidomycosis (an unapproved indication).The relationship between fluconazole use and these events is unclear. Adverse fetal effects have been seen in animals only at high-dose levels associated with maternal toxicity. There were no fetal effects at 5mg/kg or 10mg/ kg; increases in fetal anatomical variants (supernumerary ribs, renal pelvis dilation) and delay in ossification were observed at 25mg/ kg and 50mg/ kg and higher doses. At doses raging from 80mg/kg (approximately 20-60 times the recommended human dose) to 320mg/ kg, embryolethality in rats were increased and fetal abnormalities included wavy ribs, cleft palate and abnormal craniofacial ossification.
Case reports describe a distinctive and a rare pattern of birth defects among infants whose mothers received high dose (400-800mg/day) fluconazole during most or all of the first trimester of pregnancy. The features seen in these infants include brachycephaly, abnormal facies, abnormal calvarial development, cleft palate, femoral bowing, thin ribs and long bones, arthrogryposis, and congenital heart disease.

Use in lactation: Fluconazole is found in human breast milk at concentrations similar to plasma, hence its use in nursing mothers is not recommended
Fluconazole is found in human breast milk at concentrations similar to plasma. Breast –feeding may be maintained after a single dose of 150mg fluconazole. Breast –feeding is not recommended after repeated use or after high-dose fluconazole.

CONTRAINDICATIONS:
Patients with known hypersensitivity to fluconazole or to related azole compounds. Co-administration of terfenadine is contraindicated in patients receiving fluconazole at multiple doses of ≥400 mg/ day based upon results of a multiple dose interaction study.

DRUG INTERACTIONS:
Anticoagulants: In an interaction study, fluconazole increased the prothrombin time after warfarin administration in healthy males. Though the magnitude of change was small (12%), careful monitoring of prothrombin time in patients receiving coumarin-type anticoagulants is recommended.
Sulfonylureas: Fluconazole has been shown to prolong the serum half-life of concomitantly administered oral sulfonylureas (chlorpropamide, glibenclamide, glipizide and tolbutamide) in healthy volunteers. Fluconazole and oral sulfonylureas may be co-administered to diabetic patients, but the possibility of a hypoglycaemic episode should be borne in mind.

Hydrochlorothiazide: In a kinetic interaction study, co-administration of multiple-dose hydrochlorothiazide to healthy volunteers receiving fluconazole increased plasma concentrations of fluconazole by 40%. An effect of this magnitude should not necessitate a change in the fluconazole dose regimen in subjects receiving concomitant diuretics, although the prescriber should bear it in mind.
Phenytoin: Concomitant administration of fluconazole and phenytoin may increase the levels of phenytoin to a clinically significant degree. If it is necessary to administer both drugs concomitantly, phenytoin levels should be monitored and the phenytoin dose adjusted to maintain therapeutic levels.
Oral Contraceptives: Two kinetic studies with a combined oral contraceptive have been performed using multiple doses of fluconazole. There were no relevant effects on either hormone level in the 50 mg fluconazole study, while at 200 mg daily, the AUCs of ethinyl estradiol and levonorgestrel were increased, 40% and 24%, respectively. Thus, multiple dose use of fluconazole at these doses is unlikely to have an effect on the efficacy of the combined oral contraceptive.
Rifampicin: Concomitant administration of fluconazole and rifampicin resulted in a 25% decrease in the AUC and 20% shorter half-life of fluconazole. In patients receiving concomitant rifampicin, an increase of the fluconazole dose should be considered.
Cyclosporin: A kinetic study in renal transplant patients found fluconazole 200 mg daily to slowly increase cyclosporin concentrations. However, in another multiple-dose study with 100 mg daily, fluconazole did not affect cyclosporin levels in patients with bone marrow transplants. Cyclosporin plasma concentration monitoring in patients receiving fluconazole is recommended.
Theophylline: In a placebo-controlled interaction study, the administration of fluconazole 200 mg for 14 days resulted in an 18% decrease in the mean plasma clearance rate of theophylline. Patients who are receiving high doses of theophylline or who are otherwise at increased risk for theophylline toxicity should be observed for signs of theophylline toxicity while receiving fluconazole, and therapy modified appropriately if signs of toxicity develop.
Terfenadine: Because of the occurrence of serious cardiac dysrhythmias secondary to prolongation of the QTc interval in patients receiving azole antifungals in conjunction with terfenadine, interaction studies have been performed. One study at a 200 mg daily dose of fluconazole failed to demonstrate a prolongation in QTc interval. Another study at a 400 and 800 mg daily dose of fluconazole demonstrated that fluconazole taken in doses of ≥400 mg/day significantly increases plasma levels of terfenadine when taken concomitantly. The combined use of fluconazole at doses of ≥400 mg with terfenadine is contraindicated (see Contraindications). The co-administration of fluconazole at doses <400 mg/day with terfenadine should be carefully monitored.
Cisapride: There have been reports of cardiac events including torsade de pointes in patients to whom fluconazole and cisapride were coadministered.
Rifabutin: There have been reports that an interaction exists when fluconazole is administered concomitantly with rifabutin, leading to increased serum levels of rifabutin. There have been reports of uveitis in patients to whom fluconazole and rifabutin were co-administered. Patients receiving rifabutin and fluconazole concomitantly should be carefully monitored.